

Notes: Dahlquist & Ibert (JFE, 2026)

Institutions' Return Expectations across Assets and Time

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1 Big picture

- **Question.** How do institutional investors form return expectations across asset classes and over time? Are their expectations extrapolative like many retail-investor surveys, or do they move with objective, countercyclical risk premia?
- **Main data.** The paper assembles forecasts from asset managers, investment consultants, wealth advisors, public pension funds, and professional forecasters. The main asset classes are US equities, developed markets ex-US equities, emerging markets equities, US cash, and US corporate credit.
- **Core time-series result.** Subjective risk premia move nearly one-for-one with real-time objective risk premia. The estimated slopes are generally positive, close to one, and rarely statistically different from one.
- **Contrast with retail-expectations evidence.** The result differs from the literature on retail investors' extrapolative expectations. Institutions' risk premium expectations are broadly countercyclical because the objective benchmarks are countercyclical.
- **Cross-sectional result.** Even though subjective premia vary substantially over time, many equity-premium forecasts vary even more across institutions. Some institutions are persistently optimistic and others persistently pessimistic.
- **Source of disagreement.** Disagreement in US equity return expectations is mostly about long-run valuation repricing, not about income, inflation, or earnings growth. Institutions disagree about the long-term mean of valuation ratios such as CAPE.
- **Valuation-disagreement result.** Professional forecaster disagreement is U-shaped in valuation levels: disagreement is high when valuation ratios are very low or very high.
- **Across-asset persistence.** Optimism is persistent across asset classes. Institutions that are optimistic about one asset class tend to be optimistic about others.
- **Economic takeaway.** Institutional expectations look rational in their time-series comovement with objective risk premia, but they are not full-information identical expectations. The key friction is persistent heterogeneity in long-run valuation beliefs.

2 Data and measurement

2.1 Forecaster groups

The paper studies five groups of forecasters.

- **Asset managers:** large global asset managers reporting capital market assumptions.
- **Investment consultants:** institutions advising pension funds and other asset owners.
- **Wealth advisors:** advisory firms reporting long-run expected returns for portfolio construction.
- **Public pension funds:** US public pension plans reporting expected asset-class returns.
- **Professional forecasters:** the Survey of Professional Forecasters (SPF) and Livingston survey.

Interpretation. The sample is designed to capture expectations that plausibly matter for asset allocation. Investment consultants are especially important because their direct discretionary AUM can understate the assets influenced by their advice.

2.2 Subjective risk premia

The central object is a subjective risk premium expectation:

$$F_{i,t}[r_{t \rightarrow t+h}^e],$$

where forecaster i reports an expected excess return for asset class e at date t over horizon h .

- For equities, the paper studies US equity, developed markets ex-US equity, and emerging markets equity premia.
- For cash, the subjective cash risk premium is the expected long-term cash return minus the relevant Treasury-yield benchmark.
- For credit, the subjective credit risk premium is the expected corporate bond return over the cash or bond benchmark.
- Forecast horizons are generally long-run horizons. The Livingston survey is a shorter-horizon exception and is matched to an option-implied one-year benchmark.

Interpretation. The paper focuses on risk premia rather than raw returns, because it asks whether expected compensation for risk moves with objective risk premia.

2.3 Objective risk premia

The paper constructs objective risk premium benchmarks that are available in real time.

For long-term equity premia, the benchmark is based on a present-value/CAPE relation:

$$E_t[r_{t \rightarrow t+10}^e] = \ln \left(1 + \frac{1}{\text{CAPE}_t} \right) - r_{t \rightarrow t+10}^f. \quad (1)$$

For the Livingston one-year US equity forecasts, the paper uses a one-year option-implied equity premium. For cash, the objective risk premium is the negative of the Kim–Wright term premium. For credit, it is the Gilchrist–Zakrajsek excess bond premium.

Interpretation. The benchmark is intentionally simple. The goal is not to build the best return predictor, but to capture a transparent, real-time, model-based risk premium that institutions could plausibly use.

2.4 AUM weights

Some analyses weight institutions by discretionary assets under management.

- AUM is obtained mainly from Form ADV for asset managers, investment consultants, and wealth advisors.
- Pension fund AUM is obtained from Public Plans Data.
- AUM-weighted regressions ask whether the expectations of larger institutions move with objective premia.

Interpretation. Equal-weighted results describe the average institution. AUM-weighted results move closer to the expectations that may influence aggregate asset demand.

2.5 Building blocks of equity returns

To study why institutions disagree, the paper decomposes expected nominal equity returns into:

$$r_{t+1} = dp_{t+1} + \Delta e_{t+1} + \pi_{t+1} + \Delta pe_{t+1}, \quad (2)$$

$$= \text{income} + \text{real earnings growth} + \text{inflation} + \text{repricing}. \quad (3)$$

Taking expectations gives

$$E_t(r_{t+1}) = E_t(dp_{t+1}) + E_t(\Delta e_{t+1}) + E_t(\pi_{t+1}) + E_t(\Delta pe_{t+1}). \quad (4)$$

Interpretation. The key disagreement is in $E_t(\Delta pe_{t+1})$, the expected change in valuation ratios. Some institutions expect valuations to mean revert strongly; others expect little or no valuation repricing.

3 Models and empirical specifications

3.1 Subjective versus objective risk premium regression

The main regression is

$$F_{i,t}[r_{t \rightarrow t+h}^e] = a_i + bE_t[r_{t \rightarrow t+h}^e] + \varepsilon_{i,t}. \quad (5)$$

- $F_{i,t}[r^e]$ is the subjective risk premium forecast.
- $E_t[r^e]$ is the objective real-time benchmark risk premium.
- a_i is a forecaster fixed effect.
- b measures how strongly subjective expectations move with objective risk premia.

Interpretation. If $b = 0$, subjective premia do not move with objective risk premia. If $b = 1$, subjective premia move one-for-one with objective premia. If $b > 1$, forecasters overreact relative to the benchmark; if $b < 1$, they underreact.

3.2 AUM-weighted version

The AUM-weighted regression uses the same specification as Eq. (5), but weights each observation by the institution's assets under management:

$$\min_{a_i, b} \sum_{i,t} w_i (F_{i,t}[r^e] - a_i - bE_t[r^e])^2, \quad w_i \propto \text{AUM}_i. \quad (6)$$

Interpretation. This test checks whether the one-for-one relation holds for the institutions whose forecasts may matter most for aggregate portfolio allocation.

3.3 Time-series plots with forecaster and time effects

To plot subjective and objective premia over time, the paper first estimates

$$F_{i,t}[r_{t \rightarrow t+h}^e] = a_i + c_m + \eta_{i,t}, \quad (7)$$

where c_m is a month fixed effect. The fitted values net out permanent forecaster-level optimism or pessimism.

Interpretation. The plots ask whether the time-series component of subjective risk premia co-moves with objective risk premia after removing persistent differences across forecasters.

3.4 Variance decomposition

Taking variances in Eq. (7) yields

$$1 = \text{fracCov}(F_{i,t}, a_i) \text{Var}(F_{i,t}) + \frac{\text{Cov}(F_{i,t}, c_m)}{\text{Var}(F_{i,t})} + \frac{\text{Cov}(F_{i,t}, \eta_{i,t})}{\text{Var}(F_{i,t})}. \quad (8)$$

Interpretation. The first component is explained by persistent forecaster fixed effects; the second by time effects; the third is residual. If institutions only differed because they forecast at different times, time effects would dominate. If they are persistently optimistic or pessimistic, forecaster effects dominate.

3.5 Building-block variance decomposition

For US equity returns, the paper decomposes variation in expected total returns into income and growth, inflation, and repricing components. Schematically,

$$\text{Var}(E_i[r]) = \text{Cov}(E_i[r], E_i[dp + \Delta e]) + \text{Cov}(E_i[r], E_i[\pi]) + \text{Cov}(E_i[r], E_i[\Delta pe]). \quad (9)$$

Interpretation. If repricing explains most of the variance, then disagreement is mainly about long-run valuations rather than about cash-flow growth or inflation.

3.6 Valuation and disagreement regression

For professional forecaster disagreement, the paper estimates a piecewise-linear relation:

$$D_t = \alpha + \beta \log(\text{CAPE}_t) + \gamma \mathbf{1}\{\text{CAPE}_t > \overline{\text{CAPE}}\} + \delta \log(\text{CAPE}_t) \mathbf{1}\{\text{CAPE}_t > \overline{\text{CAPE}}\} + u_t. \quad (10)$$

Interpretation. The sign pattern produces a U-shaped relation: disagreement is high when valuation ratios are unusually low or high, and lower near average valuations.

3.7 Cross-asset optimism

The paper estimates institution fixed effects by asset class and then studies their pairwise correlations:

$$\hat{a}_i^A = \alpha + \rho \hat{a}_i^B + e_i. \quad (11)$$

Interpretation. A positive ρ means optimism or pessimism is persistent across asset classes. An institution optimistic about US equity premia tends also to be optimistic about DM equity, EM equity, cash, or credit premia.

4 Identification and empirical design

4.1 What identifies the main time-series result

- The coefficient b is identified from time-series variation in objective risk premia after removing forecaster fixed effects.
- The objective risk premia are available in real time, which reduces look-ahead concerns.
- The tests are run separately by asset class and forecaster type, so the result is not driven by one combined pooled regression.
- Confidence intervals use wild cluster bootstrap methods, usually double-clustering by forecaster and time.

Interpretation. The main time-series claim is not that institutions have identical expectations. It is that their expectations co-move with objective risk premium measures in a way that is close to one-for-one.

4.2 Why fixed effects matter

- Institutions have persistent average optimism or pessimism.
- Without forecaster fixed effects, persistent differences in levels could be confused with time-series comovement.
- The fixed-effect design separates average level differences from common time variation.

Interpretation. This is crucial because the paper’s two core findings are different: institutions co-move over time with objective premia, but disagree persistently in levels.

4.3 Why AUM-weighting matters

- Equal weighting gives every forecaster the same influence.
- AUM weighting asks whether larger institutions’ beliefs also co-move with objective premia.
- The AUM-weighted slopes remain close to one across asset classes.

Interpretation. This reduces the concern that the main results are driven by small institutions whose beliefs are less relevant for market pricing.

4.4 Why building-block evidence is informative

- Institutions often describe how they build expected returns from income, growth, inflation, and valuation repricing.
- The paper reads institutional white papers and reports to classify whether institutions use this building-block approach.
- The quantitative decomposition shows that valuation repricing explains most of the variation in US equity expected returns.

Interpretation. The disagreement result is not a generic statement that institutions disagree. It identifies the channel: long-run valuation beliefs.

4.5 Main limitations

- The paper observes reported expectations, not necessarily the exact beliefs used internally for every portfolio decision.
- Institutions report at different frequencies and horizons, so the authors harmonize forecasts but cannot eliminate all differences.
- Some AUM measures understate actual influence, especially for consultants.
- The objective benchmarks are deliberately simple and transparent; they are not claimed to be the best possible predictors.
- The paper does not prove whose expectations are reflected in equilibrium prices.

Interpretation. The paper is strongest as evidence on what institutions say they expect and how those stated expectations relate to real-time benchmarks and cross-sectional disagreement.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format. Adjacent tables and figures are grouped where possible to save space and speed up comparison. All visuals are cropped directly from the original paper.

5.1 Tables 1–2: sample institutions and summary statistics

Read:

- **Table 1** lists the institutions in the sample: 36 asset managers, 17 investment consultants, 11 wealth advisors, and the 20 largest public pension funds by AUM.
- Large asset managers include Vanguard, BlackRock, Fidelity, Capital Group, PIMCO, and J.P. Morgan.
- The largest consultants and advisors have direct AUM that may understate their influence because consulting mandates can affect very large pension and institutional portfolios.
- **Table 2** shows that expected equity risk premia differ across asset classes and forecaster types. Pension funds have high US equity premium expectations, while wealth advisors and asset managers often have lower averages.
- Cash premia are close to zero or negative; credit premia are generally positive.

Economic message. The data combine highly influential institutional return expectations across multiple asset classes. Table 2 already shows substantial cross-sectional heterogeneity, especially for equity premia.

Table 1
List of institutional investors.

Asset managers	AUM	Investment consultants	AUM	Wealth advisors	AUM	Pension funds (AUM top 20)	AUM
1. Vanguard	7127.7	1. Graystone Consulting	1208.0	1. Merrill Lynch	259.2	1. California PERB	373.0
2. BlackRock	4287.4	2. AON	127.3	2. Everstreet	193.7	2. California Teachers	261.3
3. Fidelity	3985.9	3. PPM	83.2	3. Morningstar	62.1	3. NY State & Local ERS	183.3
4. Capital Group	2929.0	4. Wilshire	71.2	4. Blackfield	26.5	4. Florida RS	164.9
5. PIMCO	2379.4	5. NIPC	24.7	5. Research Affiliates	13.8	5. Texas Teachers	159.4
6. J.P. Morgan	1823.4	6. Callan	21.7	6. Sun Trust	12.1	6. New York State Teachers	125.4
7. T. Rowe Price	1667.0	7. Mohr	16.8	7. Benjamin F. Edwards	5.3	7. Wisconsin RS	108.0
8. QMA	1651.0	8. Verus	6.9	8. S&S	1.8	8. Ohio PERB	90.7
9. Wells Fargo Investment Institute	1228.1	9. Angeles Investments	6.7	9. Symetric	1.3	9. Georgia Teachers	79.0
10. Invesco	1123.8	10. Clifstone	3.6	10. PDI	0.3	10. Ohio Teachers	78.2
11. Northern Trust	995.9	11. AJ Gallagher	3.0	11. CWO	0.1	11. Virginia RS	73.6
12. State Street	967.0	12. Summit Strategic Group	1.5			12. North Carolina Teachers and State Employees	72.5
13. Schwab	692.6	13. Ullwood	0.8			13. Oregon PERB	71.8
14. Franklin Templeton	680.6	14. Sellwood	0.5			14. University of California	69.4
15. MFS	553.0	15. Mercer				15. New York City ERS	67.5
16. Neuberger	549.0	16. NYW				16. Pennsylvania School Employees	59.6
17. UBS	455.3	17. CapTrust				17. LA County ERS	58.7
18. Columbia Threadneedle	414.2					18. New York City Teachers	56.5
19. BNY Mellon	368.3					19. Illinois Teachers	53.8
20. Janus Henderson	316.6					20. Michigan Public Schools	50.2
21. Voya Investment Management	308.6						
22. DWS	298.3						
23. AQR	217.5						
24. Schroders	187.9						
25. Allianz Global Investors	176.0						
26. CIBC	169.9						
27. Baitle Gifford	152.7						
28. Ninety One	148.8						
29. Arnslett Asset Management	108.8						
30. Cohen & Steers	97.2						
31. BMO	82.7						
32. GMO	80.9						
33. Pioneer Investments	71.5						
34. Robeco							
35. Royal Bank of Canada							
36. Star Capital							

The table shows the asset managers, investment consultants, wealth advisors, and the largest 20 public pension funds (out of the 214 pension funds) in the sample. The table also shows the time-series returns of diversified equity index investments (AUM) of asset managers, investment consultants and wealth advisors according to their AUM and the AUM of the 30 largest public pension funds.

(a) Table 1: institutional investors.

Table 2
Summary statistics.

Summary statistics.								
	Mean	SD	Percentile					N
			5th	25th	50th	75th	95th	
Panel A: US equity risk premium								
Asset managers	3.00	-4.40	1.24	3.02	4.24	5.64	374	
Investment consultants	3.85	1.13	2.05	3.15	3.93	4.47	5.78	133
Wealth advisors	1.68	1.85	-0.63	0.16	1.19	3.40	4.46	99
Pension funds	5.51	1.58	2.84	4.54	5.50	6.51	8.14	1309
SPF	2.99	2.14	-0.21	1.74	2.92	4.01	6.48	763
Livingston	4.27	9.02	-10.19	-0.25	4.45	8.63	18.07	1425
Panel B: DM equity risk premium								
Asset managers	3.43	2.21	-0.20	1.86	3.66	4.98	6.82	286
Investment consultants	4.59	1.21	2.65	3.89	4.49	5.28	6.78	106
Wealth advisors	4.56	1.40	2.54	3.70	4.42	5.44	7.00	98
Pension funds	6.05	1.61	3.32	5.05	6.12	7.05	8.73	525
Panel C: EM equity risk premium								
Asset managers	5.06	1.96	2.19	3.71	5.13	6.32	7.95	366
Investment consultants	5.73	1.43	3.71	4.77	5.63	6.66	8.09	123
Wealth advisors	5.81	1.73	3.15	4.77	5.62	7.03	8.95	94
Pension funds	6.46	2.20	2.93	5.10	6.59	7.74	9.72	583
Panel D: US cash risk premium								
Asset managers	-0.27	0.64	-1.49	-0.61	-0.19	0.11	0.70	311
Investment consultants	-0.48	0.72	-1.77	-0.84	-0.34	-0.05	0.44	99
Wealth advisors	-0.48	0.61	-1.41	-0.93	-0.58	-0.10	0.66	96
Pension funds	0.69	1.27	-1.41	-0.10	0.64	1.40	2.75	777
SPF	-0.77	1.20	-2.75	-1.47	-0.66	0.01	0.94	941
Panel E: US credit risk premium								
Asset managers	2.46	1.19	0.87	1.61	2.29	3.10	4.61	282
Investment consultants	2.59	1.23	0.29	1.96	2.57	3.24	4.96	95
Wealth advisors	1.96	1.36	0.02	1.06	1.78	2.91	4.50	95
Pension funds	4.24	1.46	1.66	3.25	4.13	5.21	6.79	397

The table shows means, standard deviations (SD), percentiles, and the number of observations (N) of subjective risk premia by asset class (US equities, developed markets ex-US (DM) equities, emerging markets (EM) equities, US cash, and US credit) and by forecaster-type (asset managers, investment consultants, wealth advisors, public pension funds, and professional forecasters from the SPF or the Livingston survey).

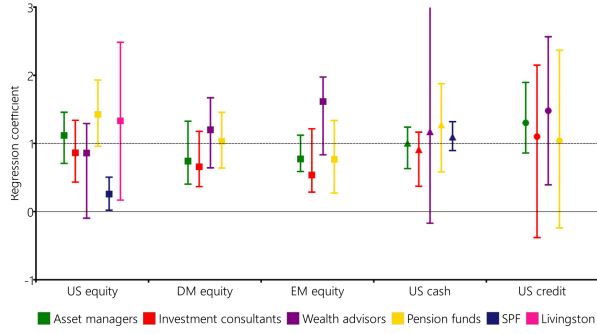
(b) Table 2: subjective risk premium summary statistics.

5.2 Figures 1–3 and Tables 3–4: subjective versus objective risk premia

Read:

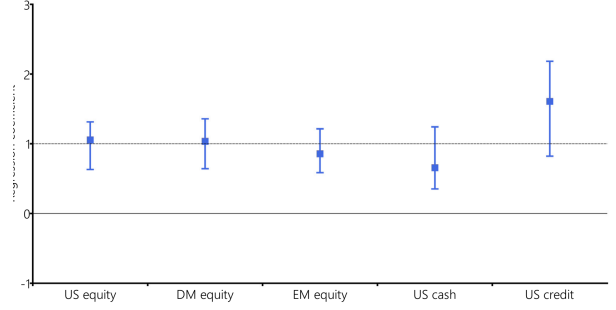
- **Table 3** reports slopes from Eq. (5). The slope for US equity is 1.117 for asset managers, 0.862 for investment consultants, 0.860 for wealth advisors, 1.427 for pension funds, 0.258 for SPF, and 1.332 for Livingston.
- Across the 23 regressions, slopes are generally positive and close to one; only one case rejects the null $b = 1$ at the 5% level.
- **Figure 1** visualizes these coefficients and confidence bands across asset classes and forecaster types.
- **Table 4** and **Figure 2** show AUM-weighted slopes: 1.055 for US equity, 1.034 for DM equity, 0.856 for EM equity, 0.656 for cash, and 1.608 for credit.
- **Figure 3** shows that objective equity and credit risk premia are countercyclical and spike around recessions, while the cash risk premium follows a different secular pattern.

Economic message. Institutions' subjective risk premia look very different from simple extrapolative retail beliefs. They move with objective, countercyclical risk premium benchmarks.



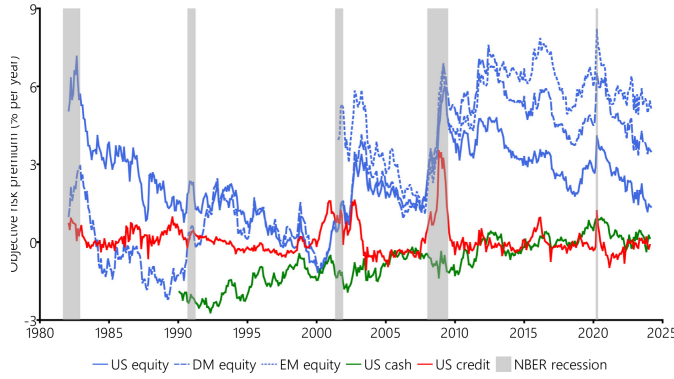
coefficients with 95% confidence bands. The figure plots the slope coefficient estimates of Table 3 together with their 95% cc

(a) Fig. 1: slope coefficients by asset and forecaster type.



coefficients with 95% confidence bands, AUM weighted. The figure plots the slope coefficient estimates of Table 4 together

(b) Fig. 2: AUM-weighted coefficients.



risk premia over time. The figure plots objective risk premia over time. Objective equity premia are based on a present-value

(a) Fig. 3: objective risk premia over time.

	b	95% confidence band		p -value of H_0		R^2		N
		Lower	Upper	$b = 0$	$b = 1$	Total	Within	
US equity	1.055	0.632	1.314	0.000	0.585	0.765	0.434	1733
DM equity	1.034	0.643	1.358	0.000	0.825	0.762	0.498	876
EM equity	0.856	0.586	1.214	0.000	0.277	0.670	0.278	1019
US cash	0.656	0.352	1.242	0.001	0.119	0.254	0.236	1124
US credit	1.608	0.822	2.184	0.003	0.147	0.409	0.331	765

The table shows the slope coefficient estimates, their 95% confidence bands, p -values for the null hypothesis of a slope coefficient equal to 0 or 1, total and within R^2 s, and the number of observations N of panel regressions of subjective risk premia on objective risk premia in which individual observations are weighted by their discretionary assets under management (ADM) as reported in Form ADV. The sample contains asset managers, investment consultants, wealth advisors, and public pension funds. The panel regressions are estimated for US equities, developed markets ex-US (DM) equities, emerging markets (EM) equities, US cash, and US credit. Objective equity premia are based on a present-value model and the regional cyclically-adjusted price-earnings ratio, the objective cash risk premium is the negative of the Kim and Wright (2005) term premium, and the objective credit risk premium is the Gilchrist and Zakrajsek (2012) excess bond premium. Confidence bands and p -values are from a wild cluster bootstrap with the null hypothesis imposed, bootstrapping by forecaster and double-clustering the variance-covariance matrix by

(b) Table 4: AUM-weighted estimates.

	$\hat{b}$	95% confidence band		p -value of H_0		R^2		N
		Lower	Upper	$b = 0$	$b = 1$	Total	Within	
Panel A: US equity risk premium								
Asset managers	1.117	0.707	1.458	0.000	0.515	0.876	0.438	369
Investment consultants	0.862	0.432	1.339	0.002	0.340	0.668	0.524	131
Wealth advisors	0.860	-0.097	1.291	0.086	0.679	0.758	0.262	99
Pension funds	1.427	0.957	1.932	0.000	0.079	0.590	0.398	1305
SPF	0.258	0.019	0.504	0.033	0.000	0.270	0.030	738
Livingston	1.332	0.169	2.485	0.024	0.540	0.372	0.174	1056
Panel B: DM equity risk premium								
Asset managers	0.741	0.402	1.326	0.001	0.568	0.771	0.365	282
Investment consultants	0.659	0.367	1.177	0.007	0.340	0.535	0.350	104
Wealth advisors	1.200	0.641	1.669	0.007	0.444	0.581	0.513	96
Pension funds	1.033	0.639	1.457	0.000	0.879	0.613	0.332	515
Panel C: EM equity risk premium								
Asset managers	0.773	0.588	1.121	0.001	0.168	0.507	0.198	356
Investment consultants	0.538	0.286	1.215	0.004	0.166	0.495	0.139	121
Wealth advisors	1.615	0.833	1.974	0.008	0.110	0.724	0.620	93
Pension funds	0.767	0.272	1.336	0.003	0.414	0.567	0.122	574
Panel D: US cash risk premium								
Asset managers	1.003	0.630	1.239	0.000	0.978	0.394	0.331	265
Investment consultants	0.909	0.371	1.167	0.001	0.579	0.415	0.290	98
Wealth advisors	1.171	-0.170	3.715	0.053	0.945	0.556	0.519	96
Pension funds	1.270	0.583	1.876	0.001	0.452	0.462	0.211	766
SPF	1.093	0.896	1.319	0.000	0.366	0.569	0.398	915
Panel E: US credit risk premium								
Asset managers	1.302	0.858	1.895	0.006	0.336	0.404	0.260	277
Investment consultants	1.101	-0.382	2.149	0.068	0.655	0.372	0.253	93
Wealth advisors	1.479	0.393	2.565	0.037	0.136	0.384	0.178	94
Pension funds	1.039	-0.239	2.367	0.127	0.949	0.417	0.064	390

The table shows the slope coefficient estimates, their 95% confidence bands, p -values for the null hypothesis of a slope coefficient equal

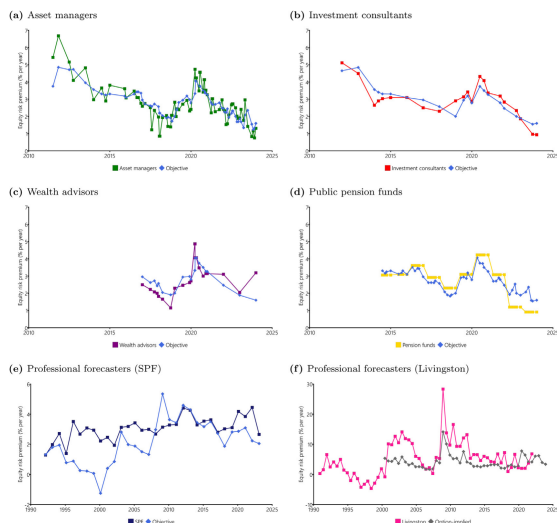
Figure 4: Table 3: subjective and objective risk premia.

5.3 Figures 4–8 and Table 5: time variation and cross-sectional disagreement

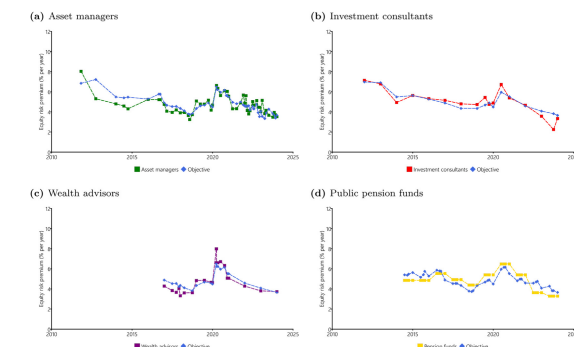
Read:

- **Figures 4–8** plot subjective risk premia and objective benchmarks over time after removing forecaster fixed effects.
- The time-series plots confirm that subjective and objective premia co-move across US equity, DM equity, EM equity, cash, and credit.
- **Table 5** shows that many equity-risk-premium expectations vary more across forecasters than over time.
- For asset managers' US equity premia, forecaster fixed effects explain 80.63% of the variation, while time fixed effects explain only 12.60%.
- For wealth advisors' US equity premia, forecaster fixed effects explain 68.57% and time fixed effects 19.38%.
- Cash and credit show less disagreement: time effects often explain more than forecaster effects.

Economic message. The paper's central tension is visible here: institutions respond to the same aggregate risk-premium signals over time, but they have persistent level differences in equity expectations.

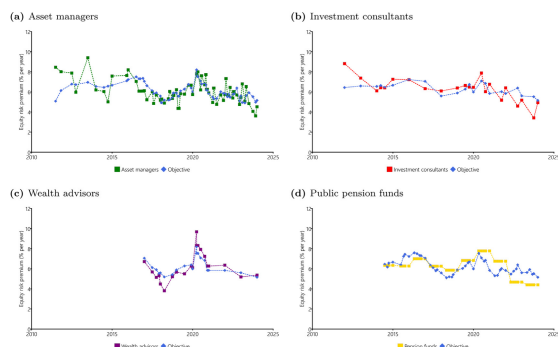


(a) Fig. 4: US equity.



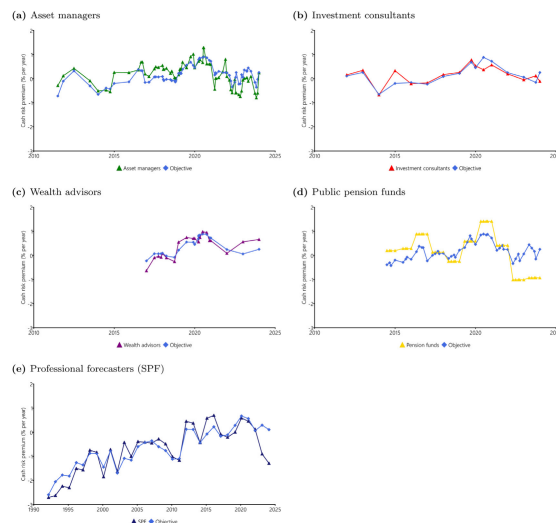
(b) Fig. 5: DM equity.

e model and the regional cyclically-adjusted price-earnings ratio.



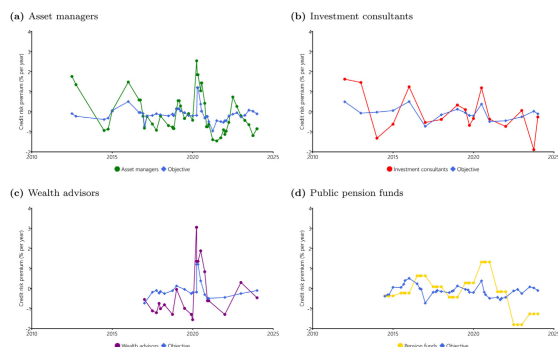
jective and objective EM equity risk premia. The figure shows subjective and objective emerging markets (EM) equity premia over time for investment consultants, wealth advisors, and public pension funds. The subjective risk premia are fitted values from a panel regression of

(a) Fig. 6: EM equity.



jective and objective US cash risk premia. The figure shows subjective and objective cash risk premia over time for asset managers,

(b) Fig. 7: US cash.



jective and objective US credit risk premia. The figure shows subjective and objective corporate bond risk premia over time for asset consultants, wealth advisors, and public pension funds. The subjective risk premia are fitted values from a panel regression of subjective r and time fixed effects, see Eq. (3). The constant is set to the mean of the objective risk premium. The time fixed effects are year-month fi

(a) Fig. 8: US credit.

Table 5

Variance decompositions of subjective risk premia.

	Standard deviation		Variance decomposition				p -value
	CS	TS	Forecaster	Time	Residual	Total	
Panel A: US equity risk premium							
Asset managers	2.93	1.04	80.63	12.60	6.77	100	0.000 ^a
Investment consultants	0.74	0.80	31.03	52.72	16.25	100	0.125
Wealth advisors	1.32	0.90	68.57	19.38	12.05	100	0.002 ^a
Pension funds	1.26	1.17	38.92	36.34	24.74	100	0.682
SPF	1.96	1.48	31.52	11.01	57.47	100	0.000 ^a
Livingston	6.49	6.07	19.98	41.45	38.57	100	0.000 ^b
Panel B: DM equity risk premium							
Asset managers	1.89	1.09	71.06	18.84	10.10	100	0.001 ^a
Investment consultants	0.86	0.87	35.89	50.40	13.71	100	0.356
Wealth advisors	0.80	0.83	22.47	61.03	16.50	100	0.133
Pension funds	1.15	1.09	41.01	40.65	18.35	100	0.961
Panel C: EM equity risk premium							
Asset managers	1.58	1.31	46.74	28.56	24.70	100	0.014 ^a
Investment consultants	1.12	0.94	51.65	34.17	14.18	100	0.293
Wealth advisors	0.97	1.12	39.09	49.46	11.45	100	0.478
Pension funds	1.62	1.26	42.65	34.12	23.23	100	0.349
Panel D: US cash risk premium							
Asset managers	0.42	0.55	21.14	42.46	36.40	100	0.020 ^b
Investment consultants	0.53	0.49	46.35	26.52	27.12	100	0.437
Wealth advisors	0.35	0.44	17.47	54.70	27.83	100	0.003 ^b
Pension funds	1.13	0.91	38.31	35.92	25.77	100	0.776
SPF	0.76	0.80	14.17	56.10	29.73	100	0.000 ^b
Panel E: US credit risk premium							
Asset managers	0.69	0.93	22.31	62.54	15.14	100	0.000 ^b
Investment consultants	1.00	0.89	28.55	45.35	26.10	100	0.096
Wealth advisors	0.92	0.89	25.97	60.42	13.61	100	0.049 ^b
Pension funds	0.87	1.00	36.12	44.49	19.39	100	0.439

(b) Table 5: variance decomposition.

5.4 Tables 6–8: building-block approach and the repricing channel

Read:

- **Table 6** reads institutions' documents to classify whether they use a building-block approach, whether they link earnings growth to GDP growth, and whether they use a time-varying target P/E ratio.
- Most institutions use a building-block approach or a close variant.
- Institutions with strongly negative repricing expectations, such as GMO, tend to have lower US equity premium expectations.
- **Table 7** shows that the average building-block components for subjective US equity returns are income and growth of 4.51%, inflation of 2.23%, repricing of -0.98%, and a total US equity return of 5.76%.
- **Table 8** shows that repricing explains most variation in US equity return expectations: roughly 68–70% across specifications.

Economic message. Disagreement about equity returns is mostly disagreement about future valuation ratios, not disagreement about dividend income, real earnings growth, or inflation.

Table 6
Building-block approach, repricing, and subjective US equity premium.

Asset managers	(1) Building block	(2) GDP growth	(3) TV target P/E ratio	(4) Re- pricing	(5) Equity premium	Investment consultants	(1) Building block	(2) GDP growth	(3) TV target P/E ratio	(4) Re- pricing	(5) Equity premium
GMO	Y	Y		-6.00	-2.91	Grayscale Consulting	Y	Y	Y	-1.43	2.45
MFS					-0.71	Summit Strategies Group	Y			-0.25	2.64
Ninety One	Y	Y	Y	-1.08	0.65	Sellwood	Y	Y		0.00	3.02
Vanguard	Y		Y	-2.42	1.87	Venus	Y	N	Y	-0.69	3.11
CIBC	Y	Y	Y	-1.50	2.11	Angeles Investments	Y	Y		-0.19	3.13
State Street	Y	Y			2.40	RVK					3.23
Star Capital					2.61	Mercor					3.28
Buillie Gifford	Y	Y		-2.10	2.70	PFM					3.49
Fidelity	Y	Y	Y	-2.42	2.73	NEPC	Y			0.51	3.81
Voya Investment Management		Y			3.06	Wilshire	Y	Y		0.25	3.87
Cohen & Steers					3.26	AJ Gallagher					4.31
Nuveen					3.26	Cliffwater	Y	Y		0.00	4.32
Capital Group	Y	Y	Y		3.28	AON	Y				4.34
J.P. Morgan	Y	Y	Y	-0.33	3.31	Ellwood	Y				4.37
Janus Henderson	N	N			3.34	Mcketa	Y	Y			4.57
Schroders	Y		Y		3.37	Callan	Y	Y		-0.15	4.58
Schwab					3.41	CapTrust	Y	Y		-0.45	5.81
PIMCO	Y			-0.22	3.42						
DWS	Y	Y		-1.14	3.45						
T. Rowe Price	Y			-0.98	3.60	Wealth advisors	(1) Building block	(2) GDP growth	(3) TV target P/E ratio	(4) Re- pricing	(5) Equity premium
Amundi Asset Management	Y	Y		-1.45	3.73						
QMA	Y	Y		-1.55	3.83	Research Affiliates	Y	Y	Y		0.54
Invesco	Y	Y	Y	-0.75	3.99	Morningstar	Y				0.68
BNY Mellon	Y	Y		-0.01	4.00	Benjamin F. Edwards	Y			0.00	2.72
UBS					4.04	Envestmet	Y	Y	Y	-0.61	3.40
AQR	Y	Y		0.00	4.05	CWO	N	N			3.73
BlackRock	Y				4.08	FIS	N	N			3.83
Alliame Global Investors	Y				4.17	Sun Trust					3.90
Northern Trust	Y	Y		-0.81	4.31	SJS					4.05
BMO					4.64	Syntronic	Y	Y		-0.01	4.17
Royal Bank of Canada					4.88	Rockefeller	Y	Y	Y	-1.10	5.03
Wells Fargo Investment Institute					4.92	Merrill Lynch					
Franklin Templeton	Y	Y	Y		4.99						
Pioneer Investments	Y				5.07						
Columbia Threadneedle	Y				5.35						
Robeco											

The table shows whether an institution uses the building-block approach of Eq. (7) ("Building block" column), whether it links its earnings growth expectations to its GDP growth expectations ("GDP growth" column), and whether its long-term target P/E-ratio is time varying ("TV target P/E ratio" column). Finally, the table shows the time-series average US repricing and US equity premium expectations for each institution (in % per year). A "Y" means "Yes", a "N" means "No", and a blank means that there is no conclusive evidence.

Figure 8: Table 6: building-block approach, repricing, and subjective US equity premia.

Table 7
Summary statistics of building blocks for subjective US equity return.

	Mean	SD	Percentile					N
			5th	25th	50th	75th	95th	
Income and growth	4.51	0.89	3.15	3.84	4.60	5.12	5.90	160
Inflation	2.23	0.31	1.75	2.00	2.20	2.45	2.70	160
Repricing	-0.98	1.21	-3.03	-1.68	-0.75	0.00	0.36	160
US equity return	5.76	1.24	3.55	5.21	5.90	6.53	7.45	160

The table shows means, standard deviations (SD), percentiles, and the number of observations (N) of building blocks for subjective US equity returns of asset managers, investment consultants, and wealth advisors. The building blocks for subjective US equity returns are income and growth, inflation, and repricing.

(a) Table 7: building-block summary statistics.

Table 8
Variance decomposition of subjective US equity return.

	Standard deviation		Variance decomposition		
	CS	TS	I	II	III
Income and growth	0.88	0.53	23.90	25.47	17.07
Inflation	0.20	0.27	8.28	5.22	13.98
Repricing	1.01	0.63	67.86	69.52	68.33
US equity return	0.93	0.76	100.00	100.00	100.00
p-value			0.001	0.018	0.003

The table shows time-series averages of cross-sectional standard deviations (CS), cross-sectional averages of time-series standard deviations (TS), and variance decompositions of subjective US equity return by asset managers, investment consultants, and wealth advisors. The variance decompositions are into the building blocks shares (in %) of income and growth, inflation, and repricing under three different specifications: I. Pooled; II. Cross-sectionally demeaned; and III. Time-series demeaned. The table also shows p-values for the null hypothesis that the variance share of income and growth is equal to the variance share of repricing. The p-values are computed using a wild cluster bootstrap with the null hypothesis imposed, bootstrapping by forecaster and double-clustering the variance-covariance matrix by month and forecaster.

(b) Table 8: variance decomposition of US equity returns.

5.5 Figures 9–10 and Table 9: valuation levels and disagreement

Read:

- **Figure 9** plots the distribution of inferred long-term CAPE beliefs for 2020–2023.
- The vertical dashed lines show realized CAPE levels. In 2021, realized CAPE was very high, and no institution expected it to rise further.
- **Figure 10** shows professional-forecaster disagreement against log CAPE. Disagreement is U-shaped: it is high when valuations are far below or far above average.

- **Table 9** estimates piecewise regressions consistent with the U-shaped relation between valuation levels and disagreement.

Economic message. When valuations are extreme, forecasters disagree about whether the market will mean revert or continue away from historical norms. This is the valuation-belief channel behind persistent disagreement.

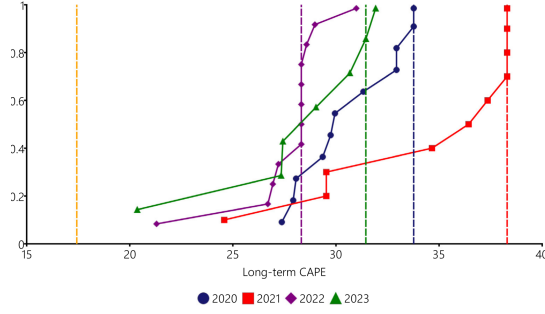
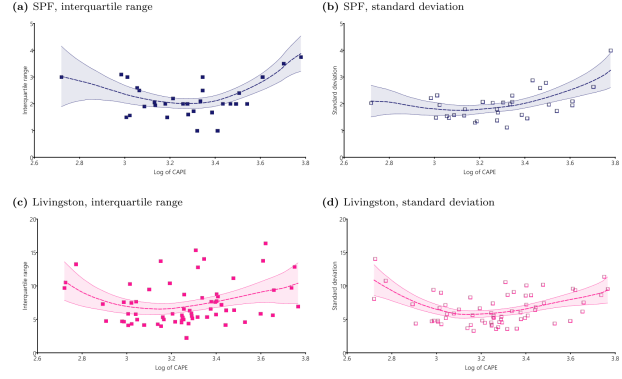


Figure 9: long-term CAPE. The figure shows the cumulative distribution functions (CDFs) of inferred subjective US long-term CAPE.

(a) Fig. 9: inferred long-term CAPE distributions.



(b) Fig. 10: disagreement and valuation levels.

Table 9
Heterogeneity in US equity risk premia and valuation levels.

	SPF		Livingston	
	Interquartile range	Standard deviation	Interquartile range	Standard deviation
Constant	8.540 (1.586)	4.008 (1.137)	26.901 (10.905)	32.547 (13.275)
Log(CAPE)	-2.056 (0.486)	-0.729 (0.363)	-6.577 (3.499)	-8.588 (4.219)
D(CAPE > Mean)	-20.300 (4.031)	-11.576 (3.923)	-42.387 (20.627)	-51.563 (17.198)
Log(CAPE) × D(CAPE > Mean)	6.100 (1.153)	3.541 (1.134)	13.397 (6.222)	16.115 (5.288)
R^2	0.000	0.000	0.110	0.000
N	33	33	67	67

The table shows regressions of disagreement measures (interquartile range and standard deviation) for the ten-year US equity premium forecasts in the SPF and the one-year US equity premium forecasts in the Livingston survey on a constant, the log cyclically-adjusted price-earnings ratio (CAPE), a dummy variable that takes on the value one if the CAPE is above its in-sample mean and that is otherwise zero, and an interaction between the log CAPE and the dummy variable. The SPF is conducted annually; the Livingston survey is conducted semi-annually. Standard errors (in parentheses) are Newey and West (1987) standard errors allowing for four lags.

Figure 11: Table 9: heterogeneity in US equity premia and valuation levels.

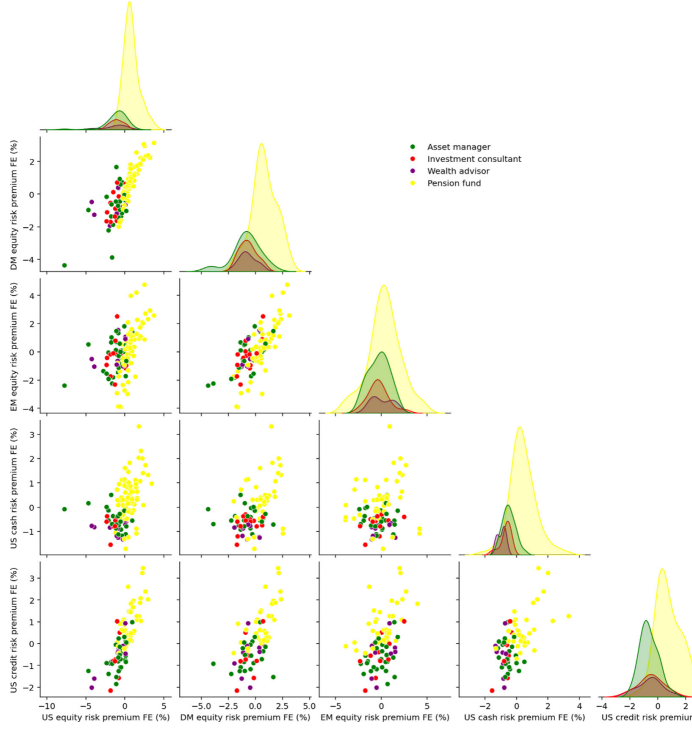
5.6 Figure 11 and Table 10: cross-asset optimism

Read:

- **Figure 11** plots pairwise relations among institution fixed effects across asset classes.
- All pairwise correlations are positive: institutions optimistic in one asset class are generally optimistic in others.
- **Table 10** quantifies this. US equity versus DM equity has a correlation of 0.806 without institution-type controls and 0.705 with them.

- US equity versus US credit has a correlation of 0.790 without controls and 0.649 with institution-type controls.
- Even after controlling for institution type, most correlations remain positive and significant.

Economic message. Persistent optimism and pessimism are not asset-class-specific quirks. They look like institution-level belief components that spill across equity, cash, and credit expectations.



(a) Fig. 11: pairwise correlations of average premia.

Table 10

Correlations across institutions.

	Correlation I		Correlation II		N
	$\hat{\rho}$	p-value	$\hat{\rho}$	p-value	
US equity vs. DM equity	0.806	0.000	0.705	0.000	124
US equity vs. EM equity	0.533	0.000	0.551	0.000	137
US equity vs. US cash	0.486	0.000	0.213	0.004	142
US equity vs. US credit	0.790	0.000	0.649	0.000	100
DM equity vs. EM equity	0.756	0.000	0.777	0.000	119
DM equity vs. US cash	0.472	0.000	0.255	0.013	88
DM equity vs. US credit	0.691	0.000	0.527	0.000	79
EM equity vs. US cash	0.221	0.000	0.215	0.083	97
EM equity vs. US credit	0.482	0.000	0.532	0.000	89
US cash vs. US credit	0.644	0.000	0.436	0.000	82

The table shows pair-wise correlations between the estimated institution fixed effect for one asset class on the estimated institution fixed effect from another asset class. Correlation I refers to the cross-sectional correlation between all institutions; Correlation II refers to the cross-sectional correlation allowing for institution-type (asset manager, investment consultant, wealth advisor, or pension fund) fixed effects. The p -values are from regressions allowing for conditional heteroskedasticity for the null hypothesis of a zero coefficient. N refers to the number of cross-sectional observations.

(b) Table 10: correlations across institutions.

6 Short summary

- The paper studies institutional and professional forecaster return expectations across equities, cash, and credit.
- The main object is the subjective risk premium: a reported expected excess return over a relevant benchmark.
- Objective equity premia are based on CAPE and real yields; objective cash and credit premia use established term-premium and excess-bond-premium measures.
- Subjective risk premia move nearly one-for-one with objective risk premia across most asset classes and forecaster types.
- AUM-weighted regressions confirm that this relation holds for large institutions.
- Time-series plots show high co-movement between fitted subjective premia and objective premia.
- Despite this time-series co-movement, equity-premium expectations show large and persistent cross-sectional disagreement.
- Forecaster fixed effects explain more variation than time fixed effects in many equity-premium panels.

- The disagreement is strongest for US equity premia and among asset managers.
- Institutions' documents show that most use a building-block approach to equity return expectations.
- The key building blocks are income, real earnings growth, inflation, and repricing.
- Repricing expectations explain most of the cross-sectional and time-series variation in subjective US equity return expectations.
- Institutions disagree mainly about long-run valuation ratios, not about inflation or cash-flow growth.
- Professional forecaster disagreement is U-shaped in CAPE: disagreement is highest when valuations are unusually low or unusually high.
- Institution-level optimism persists across asset classes.
- The paper suggests that asset prices may reflect countercyclical risk premium expectations, but the identity of the marginal belief holder remains a subject for future research.

7 Conclusion

7.1 Core conclusion

Dahlquist and Ibert show that institutional return expectations are neither simple extrapolative beliefs nor fully homogeneous rational expectations. In the time series, subjective risk premia co-move strongly with real-time objective risk premium benchmarks. In the cross-section, however, institutions disagree persistently, especially about equity premia.

7.2 Economic intuition

Institutions appear to share a broad discount-rate logic: required risk premia are high in bad times and low in good times. But they disagree about long-run valuations. Some institutions believe high valuation ratios will mean revert strongly; others believe valuation ratios can remain high or move only slowly. This difference in long-run CAPE beliefs creates persistent disagreement in equity return expectations.

7.3 Takeaway

The paper reconciles two facts that can seem contradictory. Institutional beliefs are disciplined by objective risk-premium measures over time, but they are heterogeneous across institutions. The main lesson is that institutional expectations are rational-looking in their cyclical movement yet belief-driven in their persistent cross-sectional differences.